technische universität
dortmund

Chair of Enterprise Computing
Department of Computer Science

Business Process Management

Process Identification

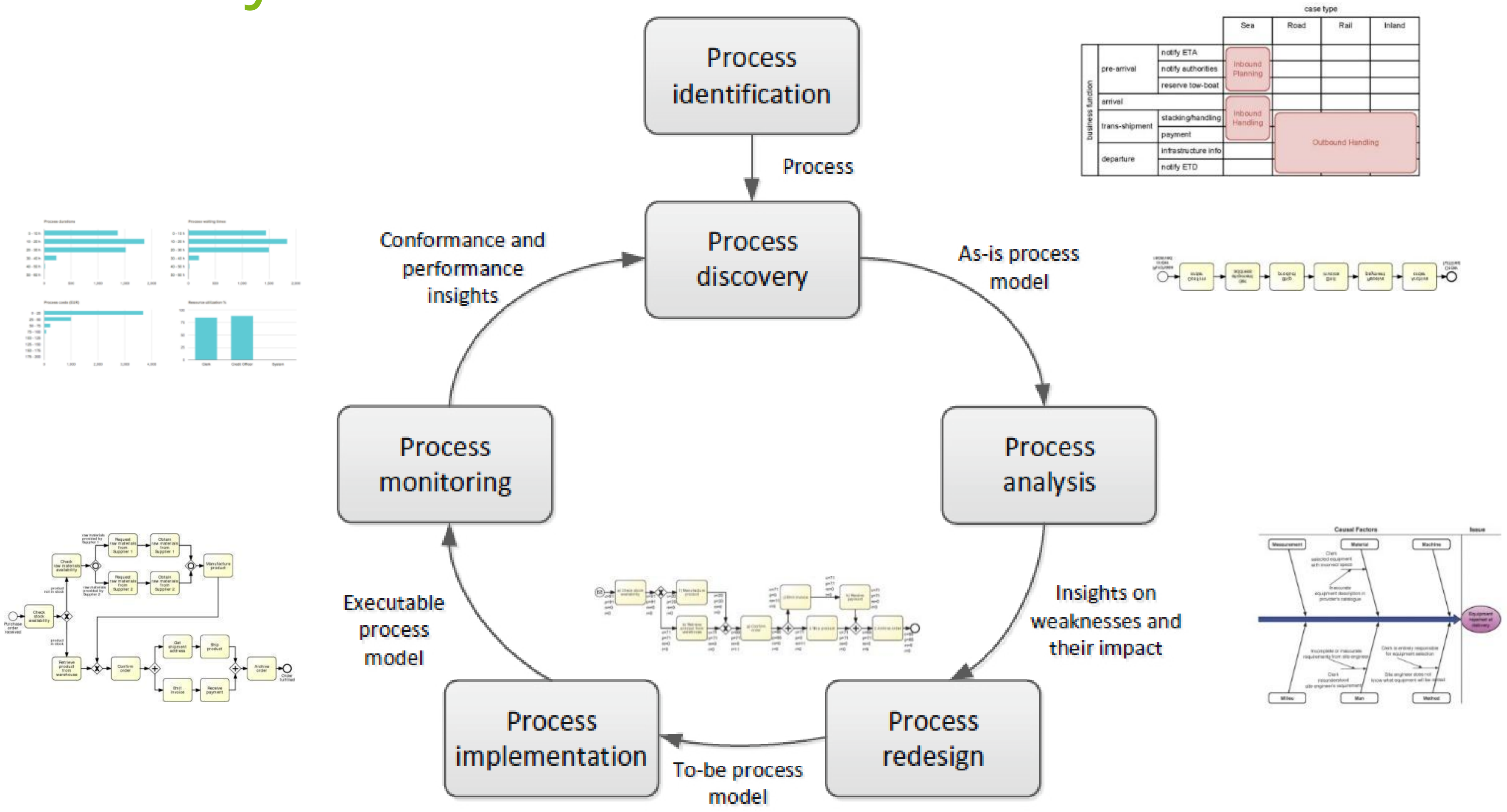
Roadmap

Introduction	10.04. Introduction	17.04. Process Identification			
Process Discovery & Modeling	24.04. Essentials of BPMN	01.05. Public Holiday	08.05. Advanced BPMN I	15.05. Advanced BPMN II	22.05. Process Discovery
Process Analysis & Redesign	29.05. Public Holiday	05.06. Qualitative Analysis	12.06. Quantitative Analysis	19.06. Public Holiday	26.06.2025 Process Redesign
Application of BPM	03.07. Process Technology	10.07. Process Monitoring and Mining	15.07. BPM as an Enterprise Capability		

Learning Goals of Today

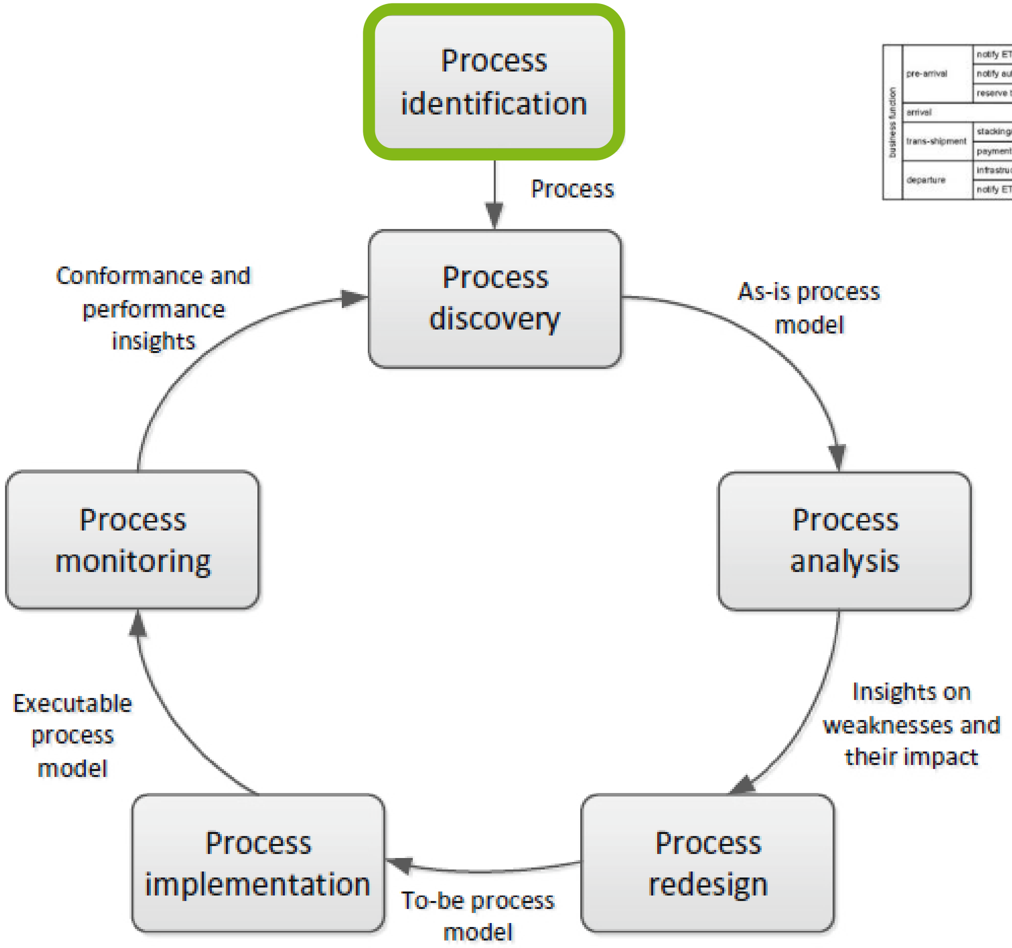
- Understand the process identification phase
- Understand process architecture definition
- Understand process selection
- Be able to work with process architectures and process portfolios

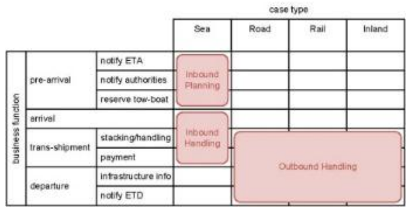
The BPM Lifecycle



Dumas et al. (2018)

The BPM Lifecycle





Dumas et al. (2018)

Overview of Process Identification




Process Identification

- ...is a **set of activities**...
- ...performed by **management, process analysts**, and **process owners**...
- ...to **establish** a process architecture (definition)...
- ...and to establish **focus** (selection) ...
- ...in order to create a working **process portfolio**.

- Can you imagine sets of processes?

- Can you imagine prioritization criteria?

Dumas et al. (2018)

Definition Phase

- **Understand** the processes and their interrelations (and the organization)
- **Define** processes and their connections
- **Determine** process scope
  - **boundaries** (horizontal and vertical) and
  - **interrelationships** (order and hierarchy)
- Caveat: There are various models to categorize processes

Davenport (1993), Dumas et al. (2018)

Selection Phase

- Establish **criteria** to prioritize the management of these processes
- **Prioritize** for modeling, redesign, automation, monitoring
- **Select** based on alignment with
  - Strategic objectives
  - Health (e.g., performance, compliance, sustainability…)
  - Feasibility to being successfully improved
  - Culture & politics
  - Risk of not improving them

Davenport (1993), Dumas et al. (2018)

Process Identification Results

- **Goal:** **Maximize value** of BPM initiatives
- **Output:** **Process Architecture** and **Process Portfolio**
- Serve as a **framework** for defining priorities and scope of subsequent BPM phases

Davenport (1993), Dumas et al. (2018)

Trade-off: Impact and Manageability

- The **larger** the process initiative, the **easier** it will become to spot opportunities for **efficiency gain**
- **But**
  - the involvement of a **large** number of staff will make effective communication **problematic**
  - keeping models of a large process **up-to-date** is more **difficult**
  - improvement projects that are related to a **large** process are more **complex**

- Rule of thumb
  - **10** to **20 main processes**
  - Distinguish **broad** and **narrow** processes

Dumas et al. (2018)

And one more thing

- Processes **change** over time
- Designing the perfect process architecture in **one go** is **unlikely**
- Identification should be **exploratory and iterative**
- Improvement opportunities are **time-constrained**

Dumas et al. (2018)

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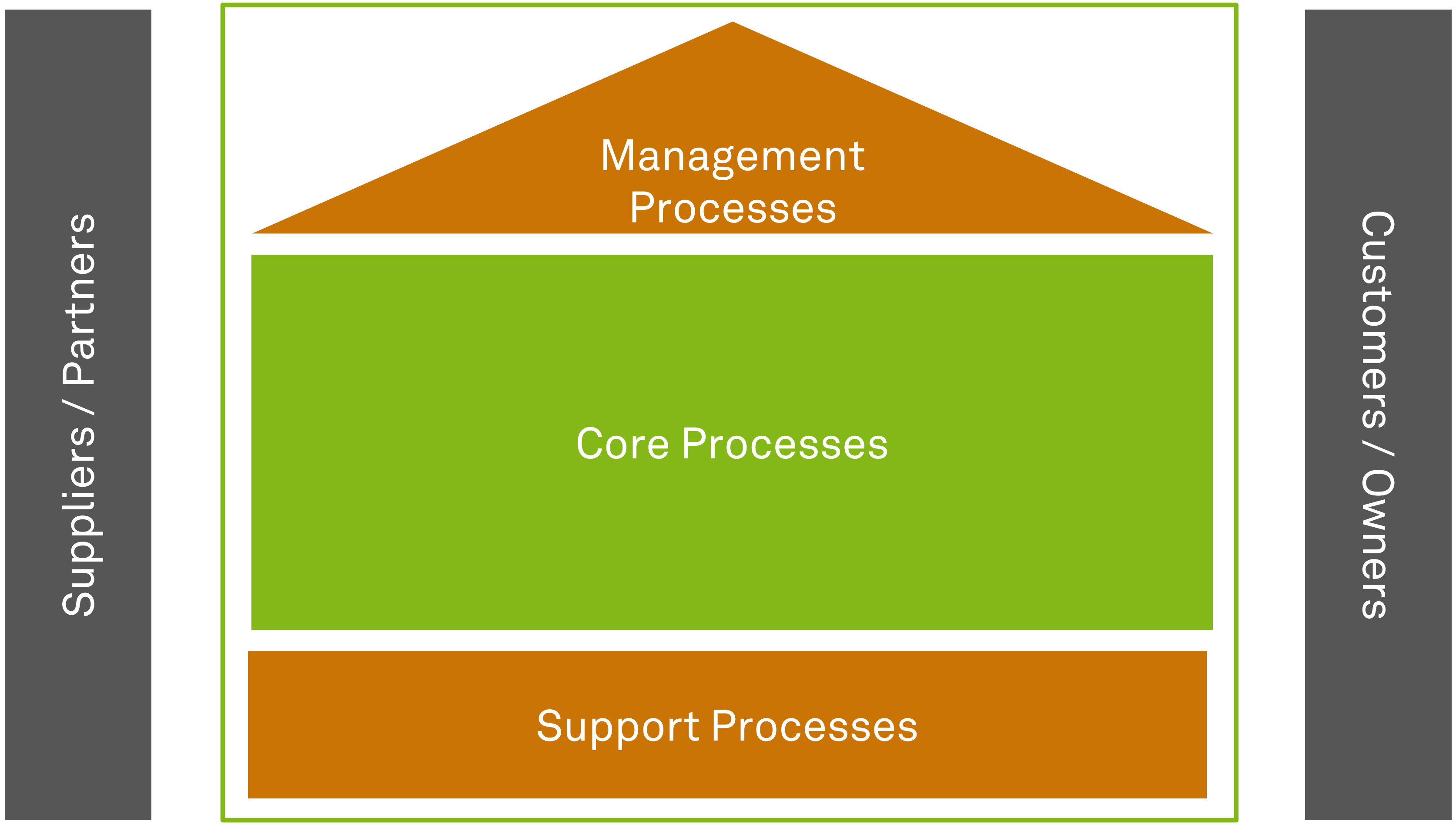
Building a Process Architecture and a Process Portfolio

Process Architecture

Process Categories



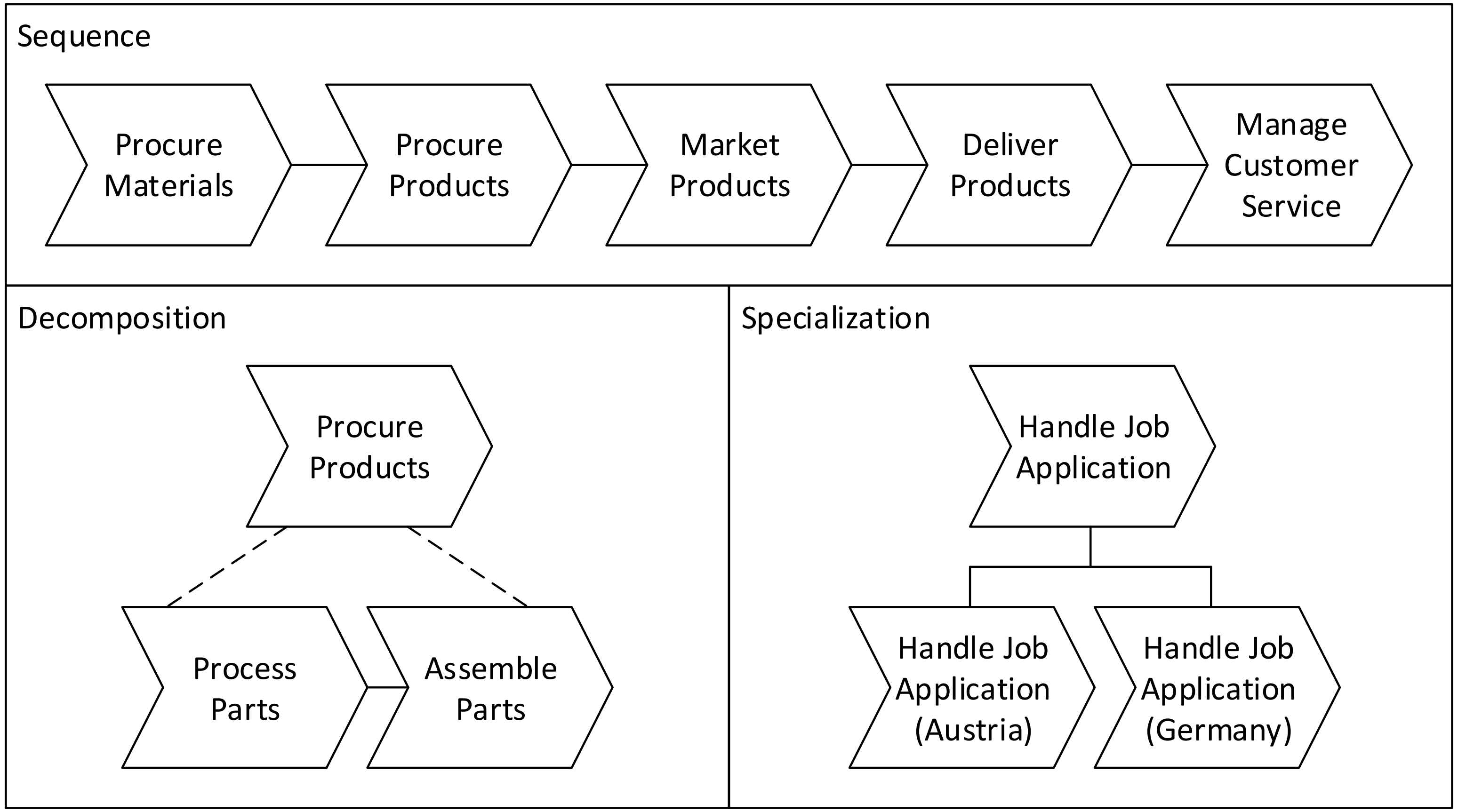
- What are core, support, and management processes at TU Dortmund University?

Porter (1985), Dumas et al. (2018)

Process Sets Illustrated – SAP Process Map

Reisert (2015)

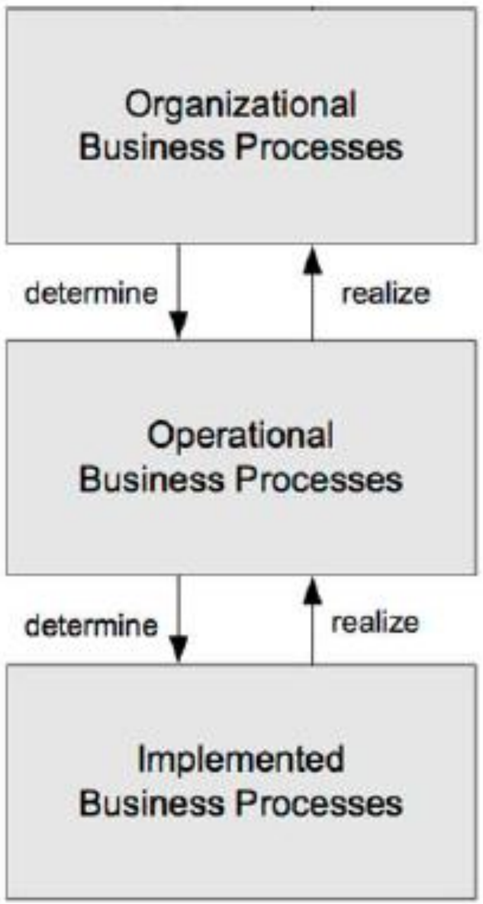
Decomposition of Business Processes (1)



Dumas et al. (2018)

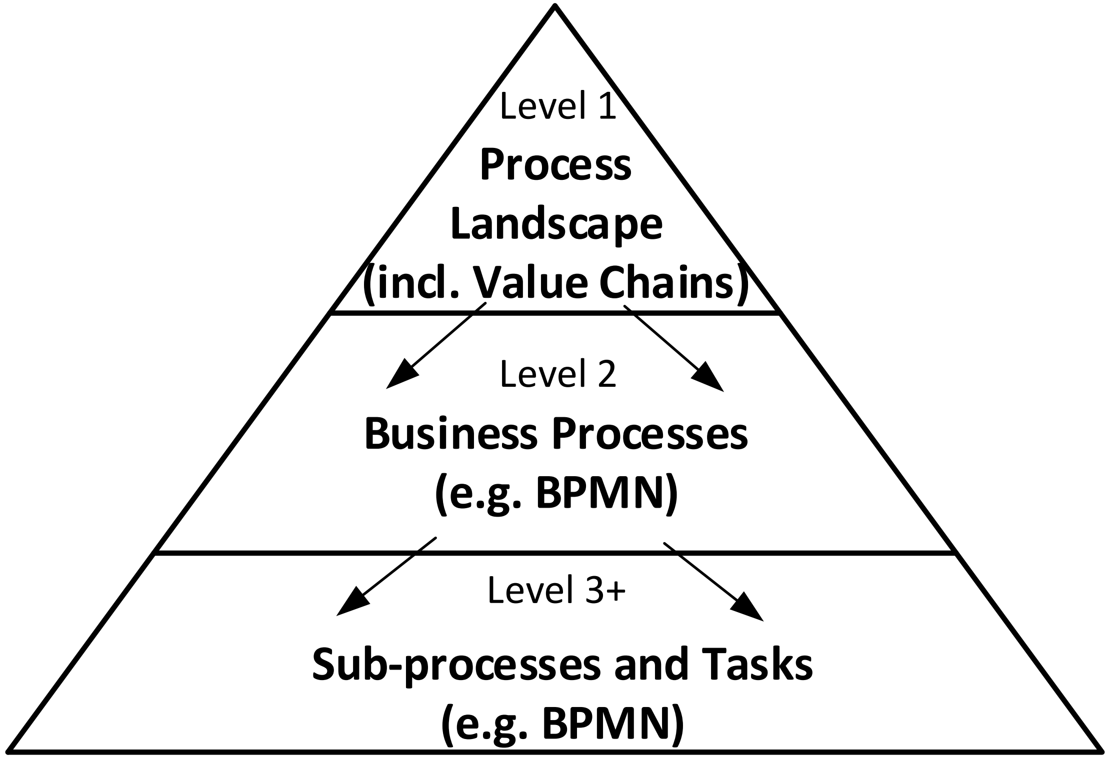
Decomposition of Business Processes (2)

- Organizational processes
  - Business processes on a **low granularity**
  - Covering the main activities of an organization
- Operational processes
  - Mapping of process activities including logical ordering
  - From the **perspective of business**
- Implemented processes
  - Include **execution information** of business processes
  - Enriched with organizational and technical aspects



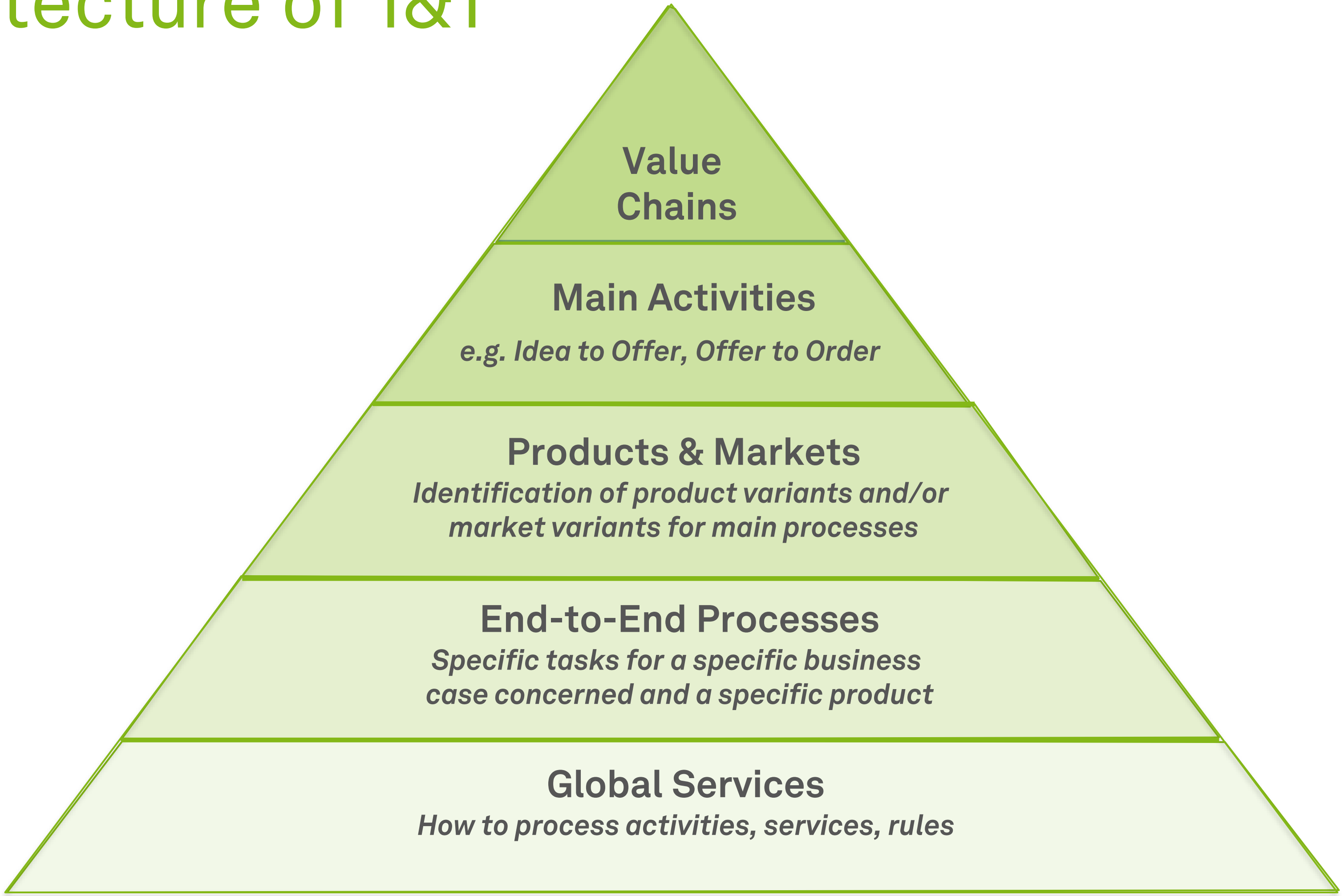
Weske (2019)

A Generic Process Architecture



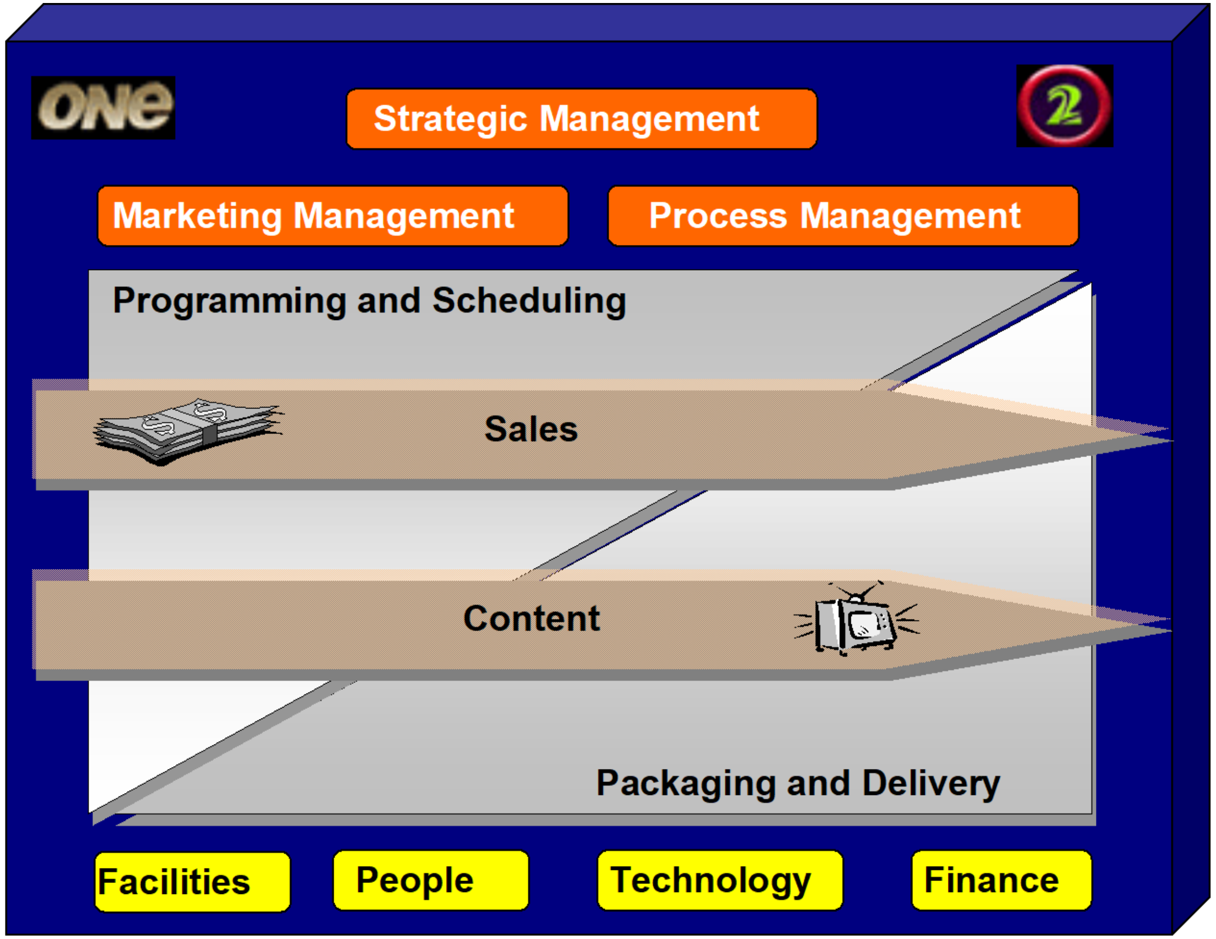
Dumas et al. (2018)

Process Architecture of 1&1



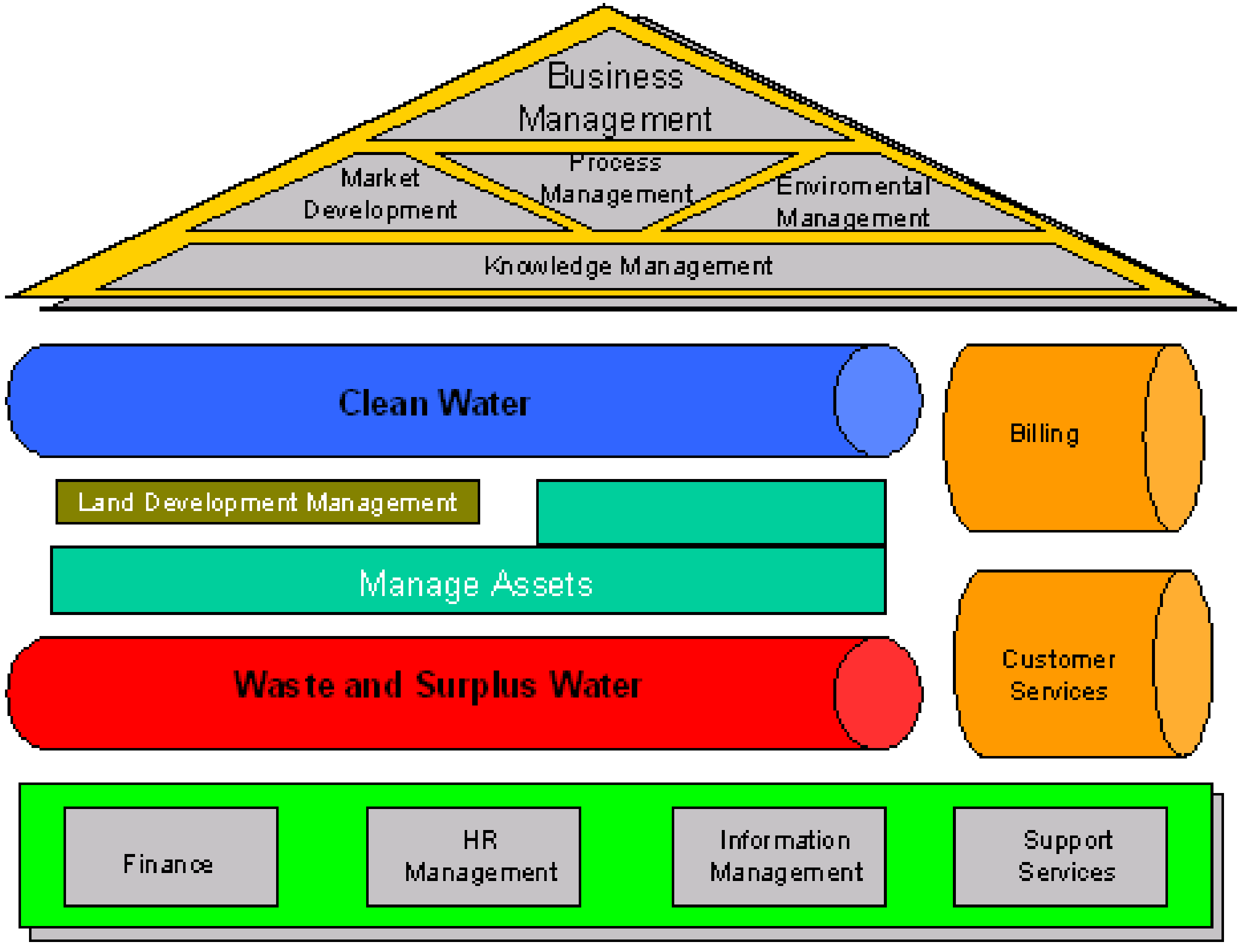
1&1 (2018)

Process Architecture of TV New Zealand



La Rosa (2013)

Process Architecture of WA Water Corporation



La Rosa (2013)

Building a Process Landscape Model (Level 1) – Overview

1. Clarify **terminology**
2. Identify **end-to-end** processes
3. For each end-to-end process, identify its **sequential processes**
4. For each business process, identify its major **management** and **support processes**
5. **Decompose and specialize** business processes
6. Compile **process profile**
7. Check **completeness** and **consistency**

Dumas et al. (2018)

Building a Process Landscape Model (1)

1. Clarify **terminology**
   – Define key terms
   – Use organizational glossary
   – Use reference models
   – Ensure that stakeholders have a consistent understanding of process landscape model
2. Identify **end-to-end** processes
   – Those processes interface with customers and suppliers
   – Goods and services that organization provides are good starting point
   – Properties help to distinguish processes, including product type, service type, channel, customer type

Dumas et al. (2018)

Building a Process Landscape Model (2)

3. For each end-to-end process, identify its **sequential** processes
   - Identify the internal, intermediate outcomes of end-to-end processes
   - Perspectives help set boundaries: product lifecycle, customer relationship, supply chain, transaction stages, change of business objects, separation
4. For each business process, identify its major **management** and **support** processes
   - What is required to execute the previously identified processes?
   - Typical support processes are management of personnel, financials, information, and materials
   - However, these can be core processes if they are integral part of business model
   - Management processes are usually generic

Dumas et al. (2018)

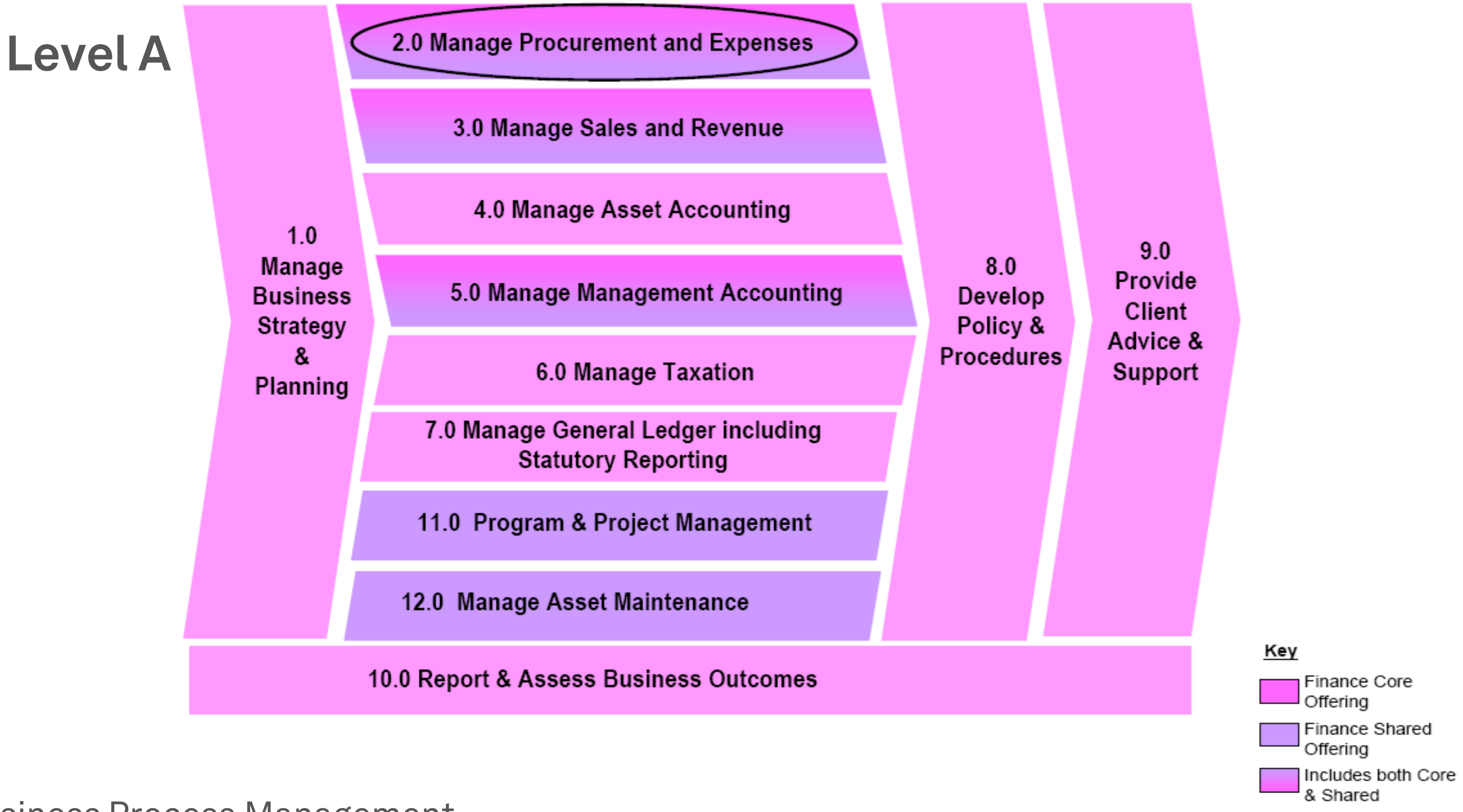
Building a Process Landscape Model (3)

5. **Decompose** and **specialize** business processes
   - Processes of Level 1 should be further subdivided into abstract process on Level 2
   - Subdivide until processes can be managed autonomously by a single process owner
   - Considerations when this subdivision should stop: Manageability and Impact
6. Compile **process profile**
   - Each of the identified processes should be described using process profile
   - Process profile supports process documentation and communication
7. Check **completeness** and **consistency**
   - Reference models can be used to check whether all major processes are included and help to check for the consistency of terminology
   - Check whether all processes can be associated with functional units of organization chart and vice versa

Dumas et al. (2018)

Process Hierarchy Example: Queensland Shared Service Agency

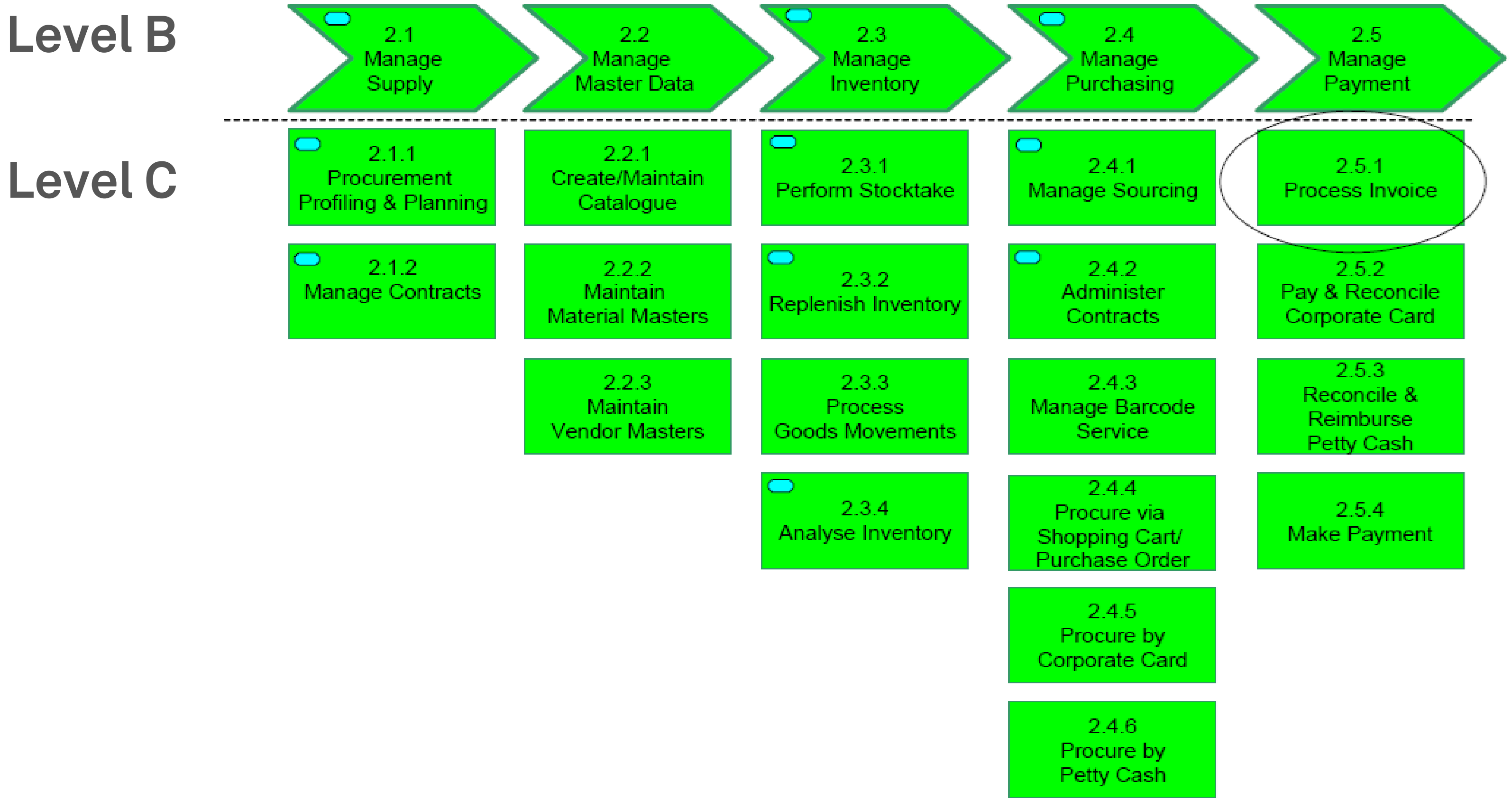
Queensland Government



La Rosa (2013)

Process Hierarchy Example: Queensland Shared Service Agency

Queensland Government

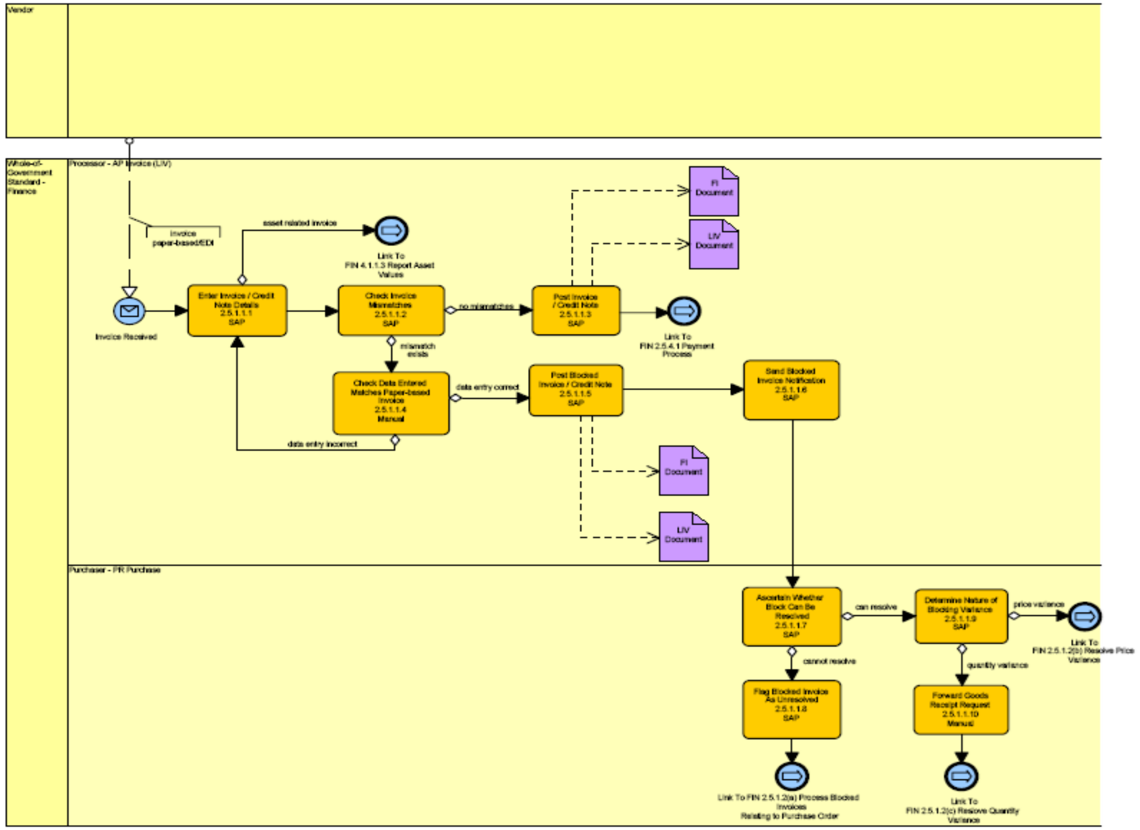


La Rosa (2013)

Process Hierarchy Example: Queensland Shared Service Agency

Queensland Government

Level D



La Rosa (2013)

Process Profile

- Accompanies a process model as a sort of fact sheet and blueprint
- Defines the boundaries (trigger, outcome) of the process
- Supports its
  - Vision
  - Process performance indicators
  - Resources, and
  - Process owner

Name of Process: Procure-to-Pay	
Vision: The objective of the procurement process is to secure that the entire range of external products and services becomes available on time and is at the required level of quality.	
Process Owner: Chief Financial Officer (CFO)	
Customer of process: • Requesting unit	**Expectation of customer:** • Timely, economic and complete provision
Outcome: Delivered products or provided services for the requested unit	
Trigger: Need is identified	
First activity: Submit Request ..... **Last activity:** Create Purchase Order	
Interfaces inbound: Plan-to-Procure **Interfaces outbound:** Construct-to-Complete	
Required resources: • Human resources: Site Engineer, Clerk, Works Engineer • Information, documents, know-how: procurement guidelines, supplier rating, framework contract • Work environment, materials, infrastructure: Procurement information system	
Process Performance Measures: • Cycle Time • Operational Costs • Error Rate	

Dumas et al. (2018)

Guiding Questions for Scoping Processes

- Where does the process **fit** into the process architecture?
  - ...if a process architecture is already in place
- On what level is the **unit of analysis**?
  - i.e., end-to-end process, procedure, or operation
- What are the **previous/subsequent processes** and what are the **interfaces** to them?
- What **variants** does this process have?
- What **underlying processes** describe elements of this process in more detail?

Dumas et al. (2018)

Excursus: Techniques for Scoping Processes

- Identify relevant **stakeholders** and **objectives**
  - e.g., via a Stakeholder-Objectives Matrix
- Identify relevant **context**
  - e.g., via a SIPOC (Suppliers, Inputs, Process, Output, Customers) Diagram
- Identify relevant **guides** and **enablers**
  - e.g., via an IGOE (Input/Guides/Enablers/Outputs) Diagram
- Identify relevant **process boundaries**
  - e.g., via a Case/Function Matrix

- A combination of the above

Dumas et al. (2018)

Excursus: Identify Relevant Process Stakeholders

- **Process owner**
  - Responsible for the effective and efficient operation of the process being modeled
- **Primary process participants**
  - Those who are directly involved in the execution of the process under analysis
- **Secondary process participants**
  - Those who are directly involved in the execution of the preceding or succeeding processes

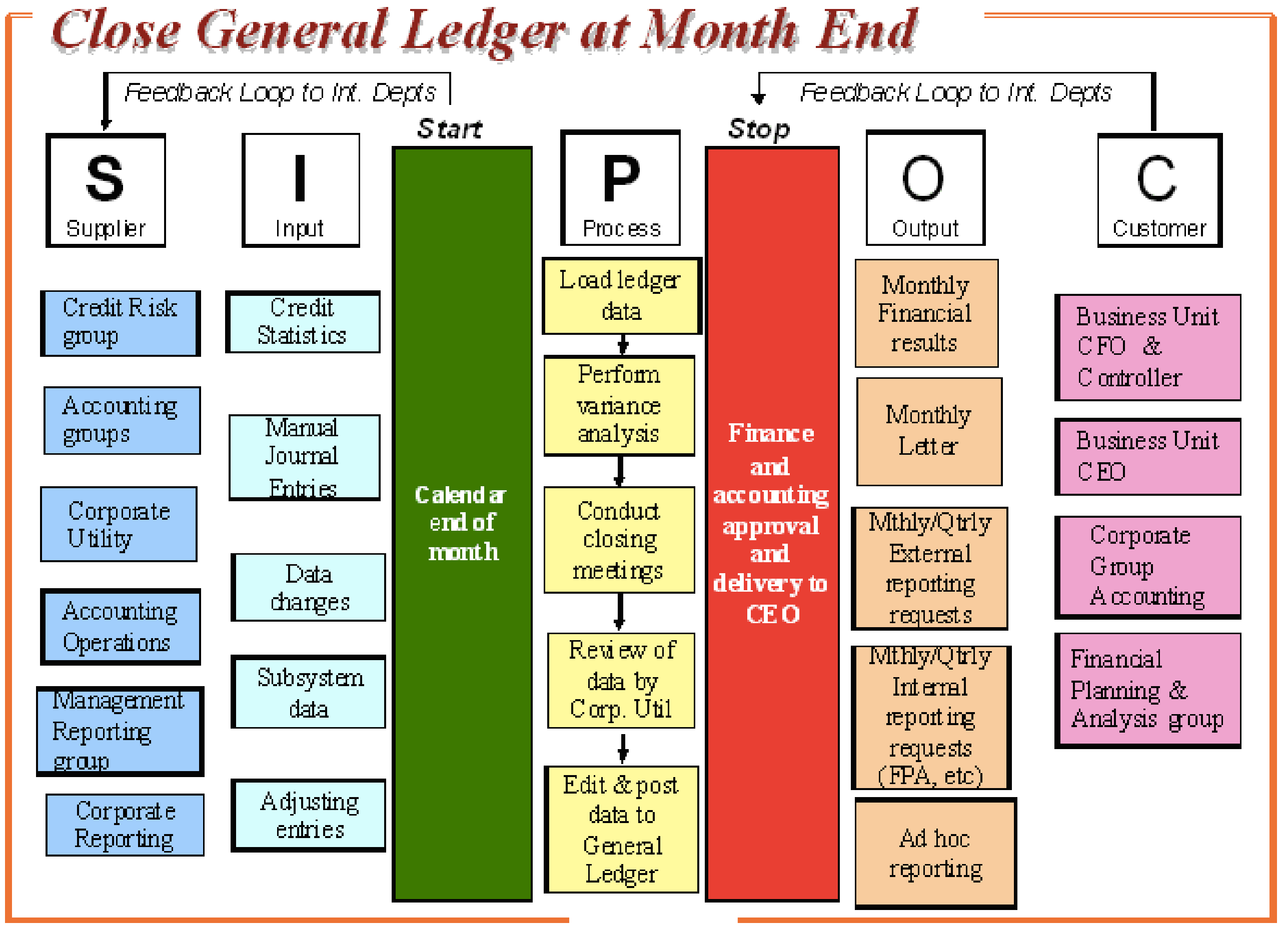
Dumas et al. (2018)

Excursus: Identify Relevant Process Objectives

- **Primary (hard)** process objectives
  - Time, cost, quality (minimize, maximize)
  - Satisfaction, compliance, flexibility, predictability
- **Secondary** process objectives
  - To purchase goods, to hire new staff members
- Back up with **appropriate process metrics**
  - e.g., cost per execution, resource utilization, waste
  - e.g., cycle time, waiting time, non-value-adding time
  - e.g., error rates, SLA violations, customer feedback
- Let involved stakeholders define their priorities

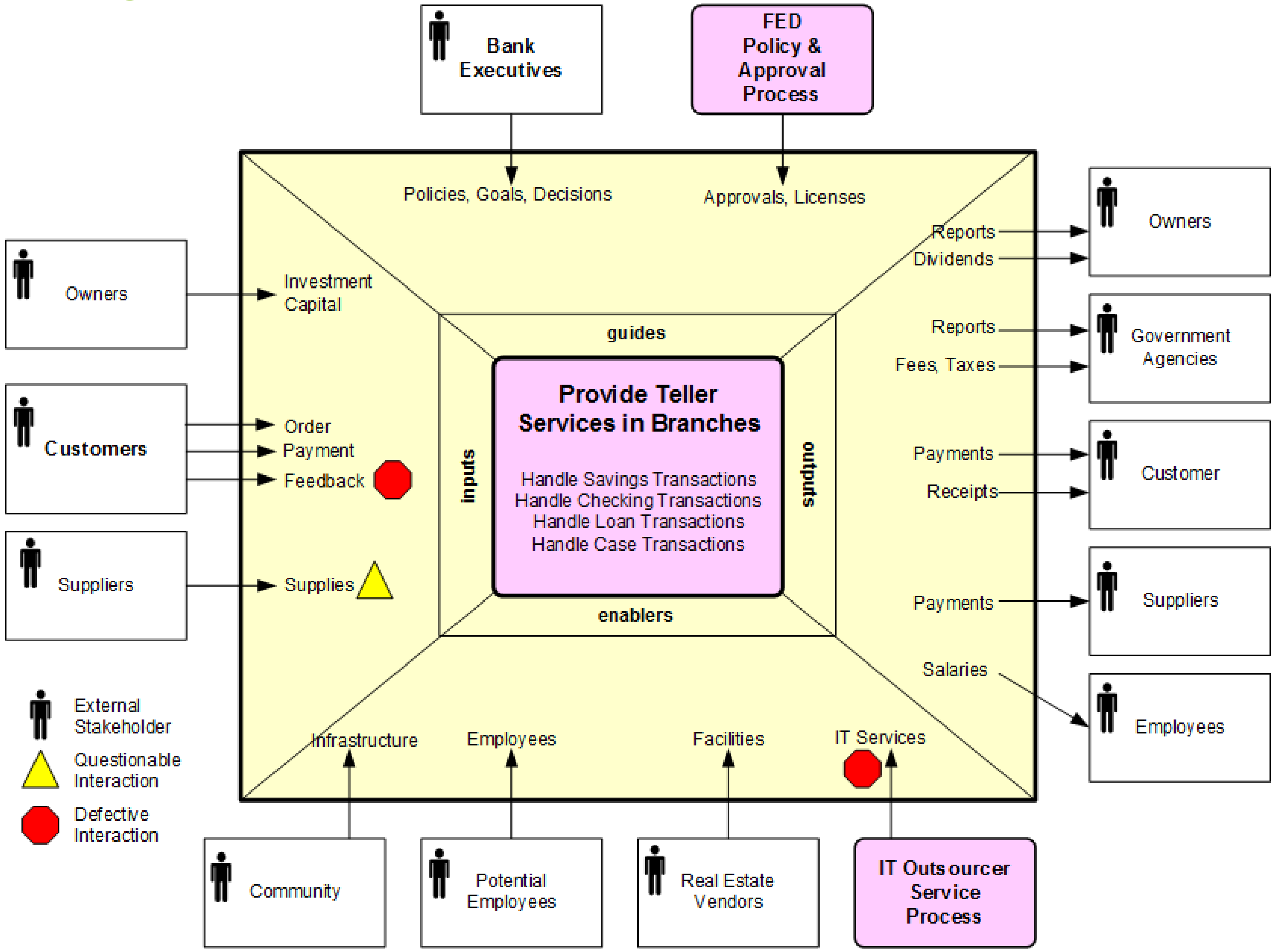
Dumas et al. (2018)

Excursus: Identify Context



American Society for Quality (ASQ) Service Quality Divison (2016)

Excursus: Identify Guides and Enablers



BPTrends (2013)

Excursus: Maturity Levels for Prioritization

- Guiding Questions
  - Does the organization cover the **expected range** of processes?
  - Are these processes **documented** and **supported**?

- Assessment is called *Appraisal*

- Example
  - Standard CMMI Appraisal Method for Process Improvement (SCAMPI)
  - Uses the Capability Maturity Model Integration (CMMI)

Dumas et al. (2013)

Excursus: CMMI Maturity Levels

- Level 1 (**Initial**)
  - Ad-hoc fashion, without any clear definition, control missing
- Level 2 (**Managed**)
  - Project planning along with project monitoring and control, measurement and analysis, process and product quality assurance
- Level 3 (**Defined**)
  - Organizations has focus on processes, process definitions available, training provided, documentation and analysis, integrated project and risk management, decision analysis and resolution
- Level 4 (**Quantitatively Managed**)
  - Process performance tracked, project management using quantitative techniques
- Level 5 (**Optimizing**)
  - Organizational performance management with causal analysis and resolution

Information Systems Audit and Control Association (ISACA) (2022)

Excursus: Reference Models

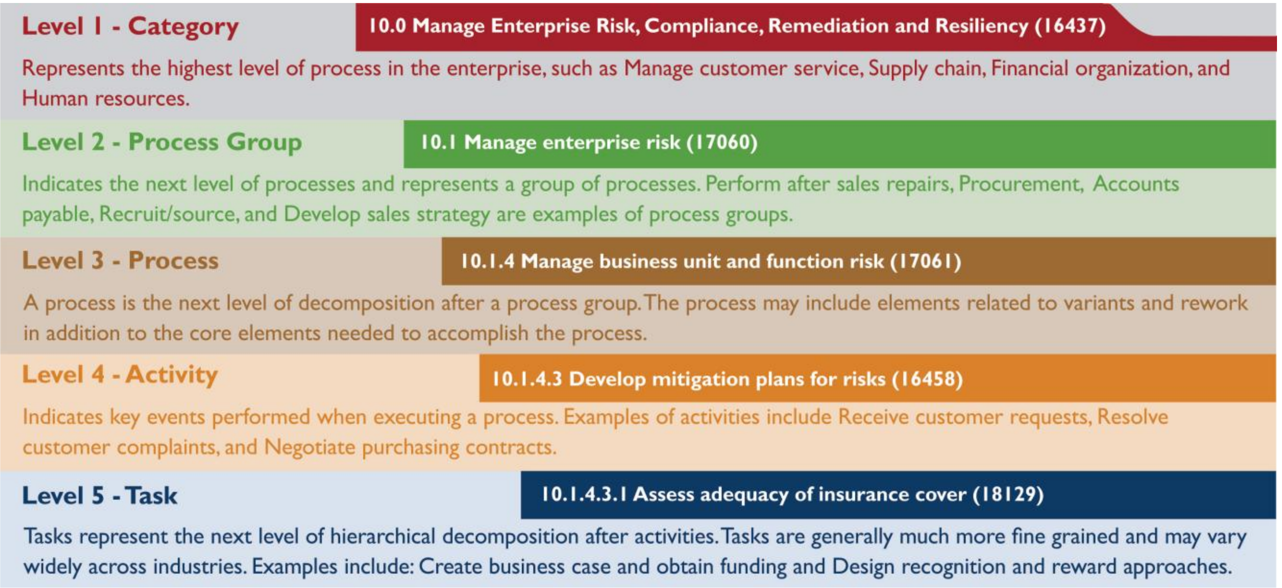
- A **reference model** is used as a template to design the process architecture

- Examples
  - Process Classification Framework (PCF)
  - Information Technology Infrastructure Library (ITIL)
  - Supply Chain Operations Reference Model (SCOR)
  - Value Reference Model (VRM)
  - Voluntary Interindustry Commerce Solutions (VICS)
  - eTOM Business Process Framework

Dumas et al. (2018)

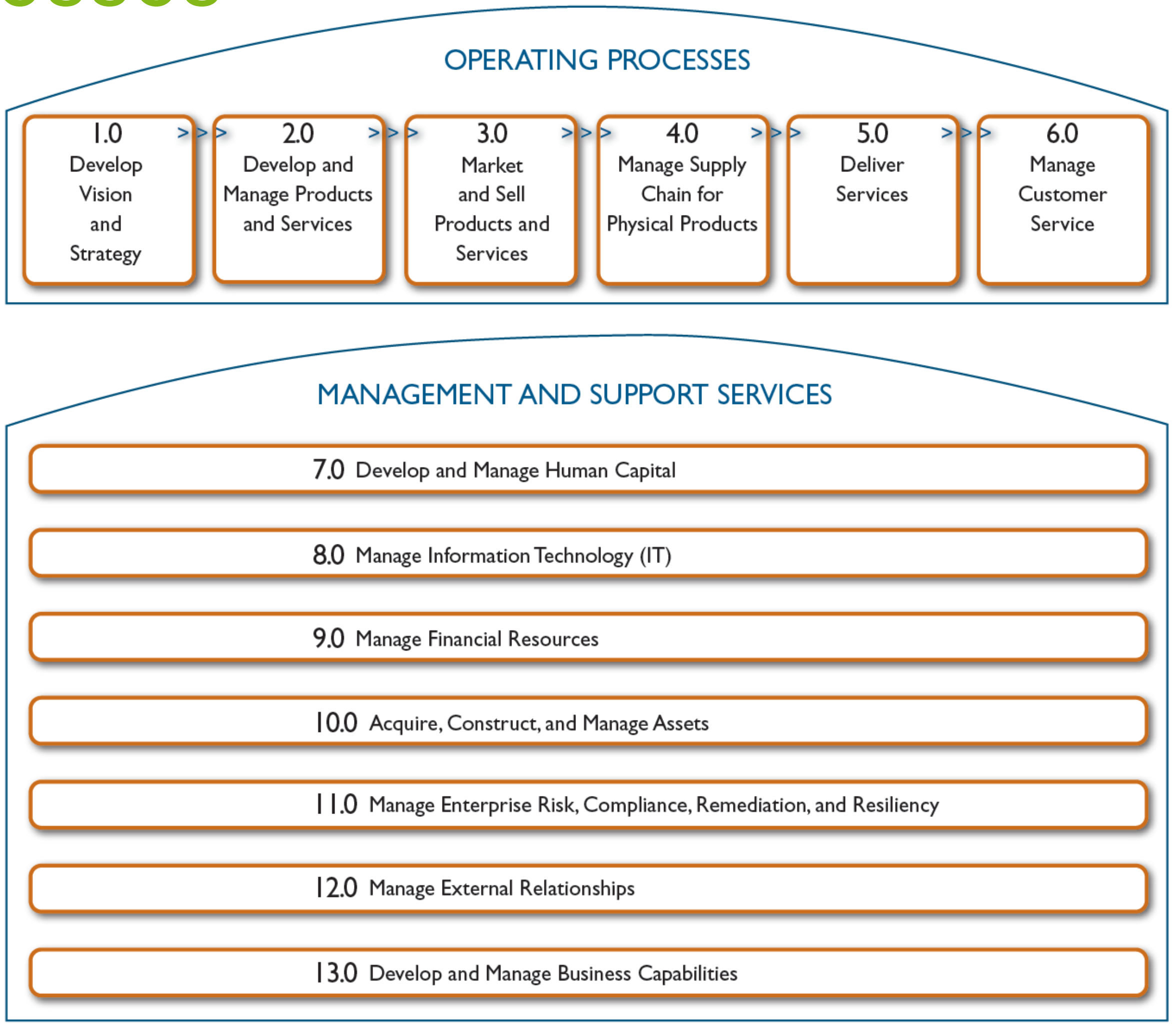
Example: Process Classification Framework (v7.3)

- PCF from American Productivity and Quality Center (APQC)
- Industry-neutral enterprise model
- Open standard for benchmarking



APQC (2022)

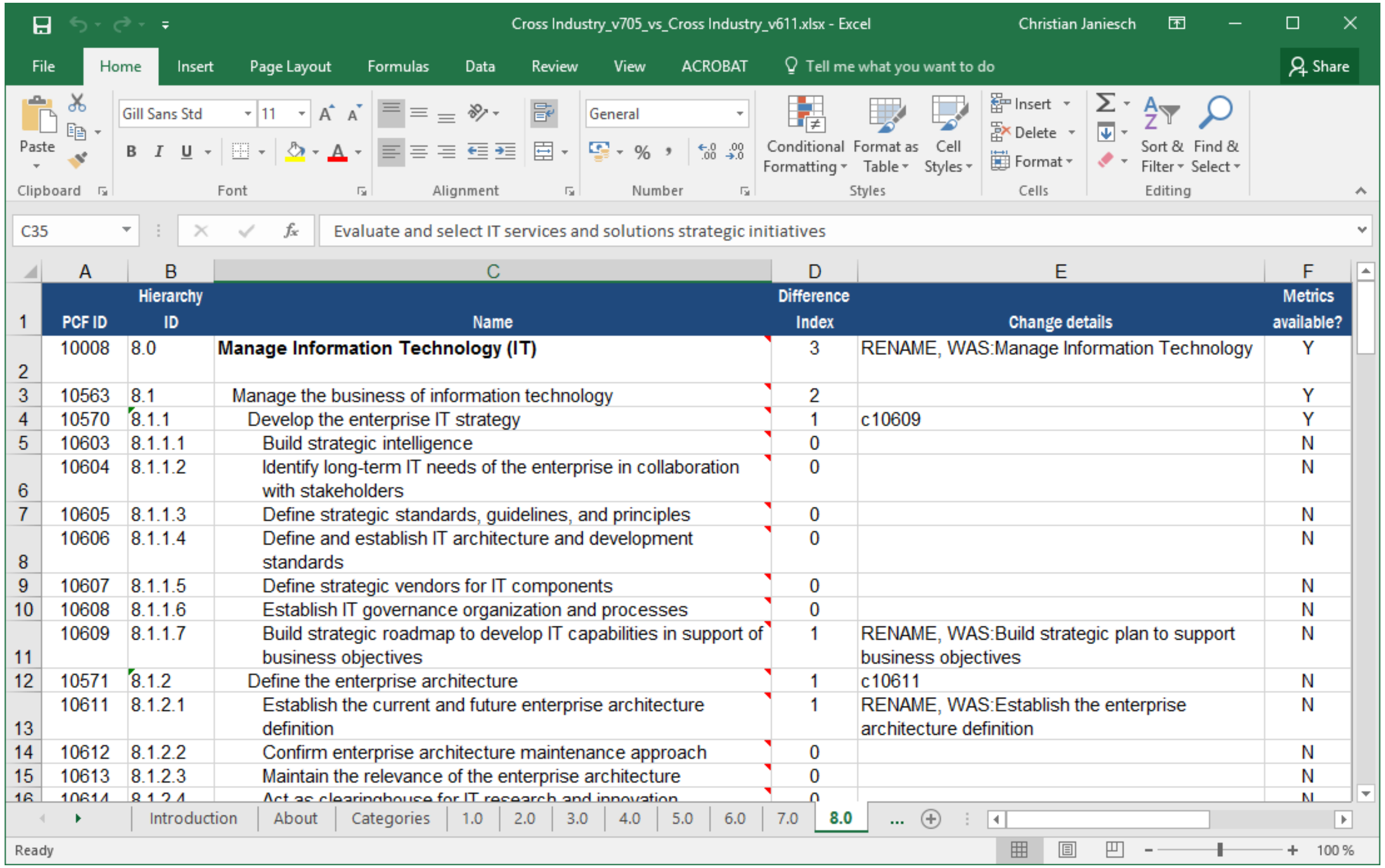
APQC PCF's Core Processes



APQC (2022)

APQC PCF's Processes

APQC American Productivity & Quality Center



PCF ID	Hierarchy ID	Name	Difference Index	Change details	Metrics available?
10008	8.0	**Manage Information Technology (IT)**	3	RENAME, WAS:Manage Information Technology	Y
10563	8.1	Manage the business of information technology	2		Y
10570	8.1.1	Develop the enterprise IT strategy	1	c10609	Y
10603	8.1.1.1	Build strategic intelligence	0		N
10604	8.1.1.2	Identify long-term IT needs of the enterprise in collaboration with stakeholders	0		N
10605	8.1.1.3	Define strategic standards, guidelines, and principles	0		N
10606	8.1.1.4	Define and establish IT architecture and development standards	0		N
10607	8.1.1.5	Define strategic vendors for IT components	0		N
10608	8.1.1.6	Establish IT governance organization and processes	0		N
10609	8.1.1.7	Build strategic roadmap to develop IT capabilities in support of business objectives	1	RENAME, WAS:Build strategic plan to support business objectives	N
10571	8.1.2	Define the enterprise architecture	1	c10611	N
10611	8.1.2.1	Establish the current and future enterprise architecture definition	1	RENAME, WAS:Establish the enterprise architecture definition	N
10612	8.1.2.2	Confirm enterprise architecture maintenance approach	0		N
10613	8.1.2.3	Maintain the relevance of the enterprise architecture	0		N
10614	8.1.2.4	Act as clearinghouse for IT research and innovation	0		N

APQC (2022)

Process Selection Criteria for Prioritization

- **Strategic Importance**
  - Which processes have the greatest impact on the organization‘s strategic goals?
  - Consider profitability, uniqueness, or contribution to competitive advantages
- **Health (Dysfunction)**
  - Which processes are in the deepest trouble?
- **Feasibility**
  - Which process is the most susceptible to successful process management?
  - Consider culture and politics!

- Result: **Process Portfolio**

Dumas et al. (2018)

Process Performance Measures for Prioritization

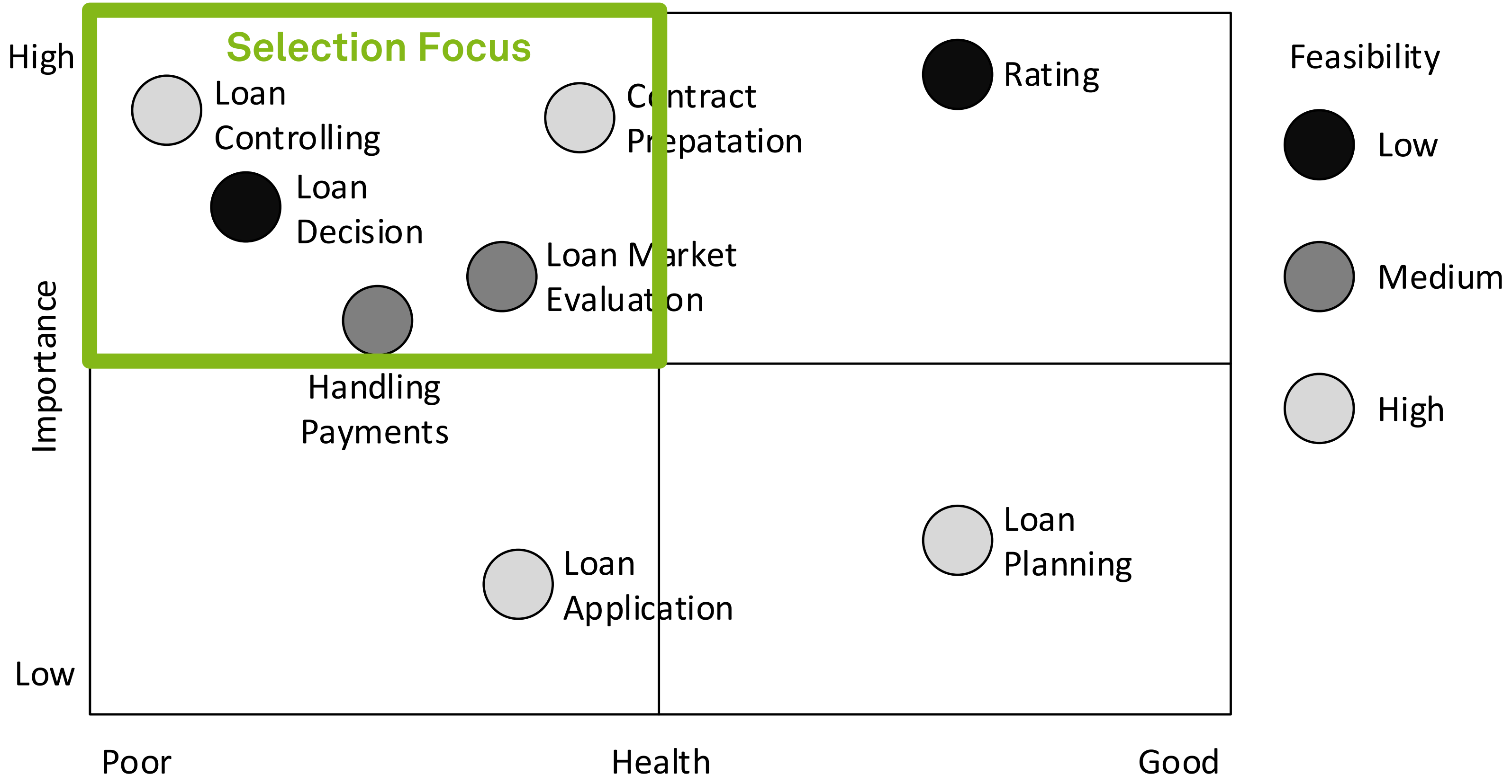
- **Time**
- **Cost**
- **Quality**
- **Flexibility**

- **Time**
- **Cost**
- **Quality**
- **Flexibility**
- **People**

- Formulate performance objectives
- Identify performance dimensions and performance measures
- Define performance target
- Result: **Process Portfolio**

Dumas et al. (2018), François et al. (2022)

Process Portfolio Example



Dumas et al. (2018)

The Hard Questions

- Does an assessment of the **importance, health,** and **feasibility** always point to the same processes to actively manage?

- Should all processes that are **not healthy**, of **strategic importance**, and **feasible** to manage be subjected to BPM initiatives?

Dumas et al. (2018)

Selection Project by Project

- Processes are identified with every **request** from a line of business
- Ensures **high relevance** for involved business unit

- **Drawbacks**
  - **Reactive** approach (-)
  - Often restricted to **discrete improvement** (-)
  - **No** conscious process **selection** approach (-)

Dumas et al. (2018)

Pitfalls of Process Identification (1)

- **Purpose** of the project is **not clear enough**
  - inappropriate scoping of the process
- **Scope** of the process is **too narrow**
  - the identified root-causes are located outside the boundaries of the process under analysis
- **Scope** of the process is **too wide**
  - process improvement project that has to be compromised in its lack of detail
- **Business process architect** has **poor facilitation skills**
  - inability to resolve emerging conflicts between the project members and stakeholders

Dumas et al. (2018)

Pitfalls of Process Identification (2)

- **Process** is **identified in isolation** to other projects due to poor portfolio management
  - redundancies and inconsistencies between projects
- Involved **project members and stakeholders** have **not** been sufficiently **informed** about the benefits of the project
  - limited participation
- Involved **project members and stakeholders** have **not** been carefully **selected**
  - very limited source of knowledge

Dumas et al. (2018)

Learning Goals of Today

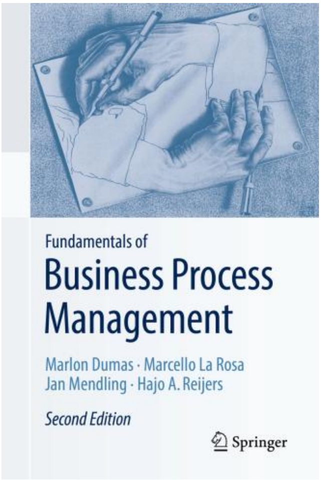
- Understand the process identification phase
- Understand process architecture definition
- Understand process selection
- Be able to work with process architectures and process portfolios

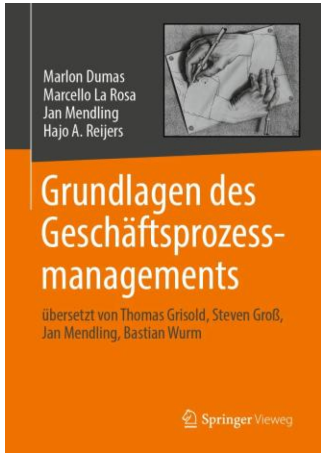
Recap Questions

- Which tasks are performed in the process identification phase and what is its outcome?
- What is the relation of a process architecture to a process landscape model?
- What is a technique for scoping business processes and what is useful for?
- What is a process portfolio and how is it related to a process architecture?

Literature

- **Section 2:** Process Identification





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Any Questions about Process Identification?

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